Autonomous Networking

Networks that observe, decide, and adapt

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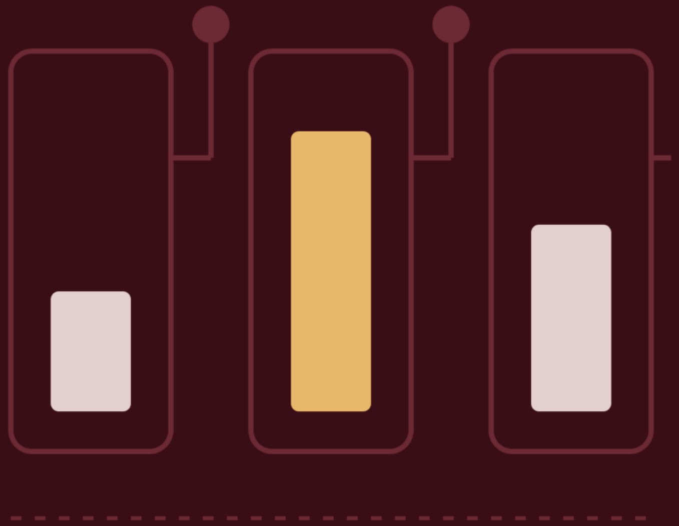
AUTONOMOUS NETWORKING · A.Y. 26-27

Reinforcement learning · lecture 3

Exploring smarter

Comparing action-selection methods on the 10-armed testbed, and upper confidence bound (UCB) exploration



RECAP · LECTURE 2

Where we left off

THE BANDIT ALGORITHM

```
$A \leftarrow \mathrm{argmax}_a\, Q(a)$   w.p. $1 - \varepsilon$
    random action w.p. $\varepsilon$
$R \leftarrow \mathrm{bandit}(A)$
$Q(A) \leftarrow Q(A) + \alpha \cdot [R - Q(A)]$
```

Q_1 · initial value

0 = neutral · high = optimistic

ε · exploration

0 = greedy · small = keeps exploring

α · step size

1/n = sample average · constant = tracks changes

Answer 1 · ε-greedy

Explore a little, forever: random action with probability ε

Answer 2 · optimism

Explore a lot, at the start: high Q_1, then greedy

Open question: which answer works better, and by how much? One small example cannot tell us.

TODAY'S QUESTIONS

Which way of exploring works best, and can we explore smarter than at random?

PART 1 · MEASURE

The **10-armed testbed**: a fair experiment to compare greedy, ε-greedy and optimistic initial values

PART 2 · EXPLORE SMARTER

Prefer the actions we are **uncertain** about: **upper confidence bound** (UCB) action selection

TODAY'S PLAN

Two parts: measure, then improve

Part 1 · Measure

→ **The 10-armed testbed**

○ ε-greedy on the testbed

○ Optimistic initial values on the testbed

Part 2 · Explore smarter

○ Uncertainty in the estimates

○ Upper confidence bound (UCB)

○ UCB on the testbed, and in networks

All the methods we compare are versions of the same algorithm: only the **selection rule** changes.

PART 1 · MEASURE

The 10-armed testbed

1 **Draw the problem**

For each of the $k = 10$ actions: $q^*(a) \sim N(0, 1)$

2 **Draw every reward**

When a is selected: $R \sim N(q^*(a), 1)$

3 **The optimal action**

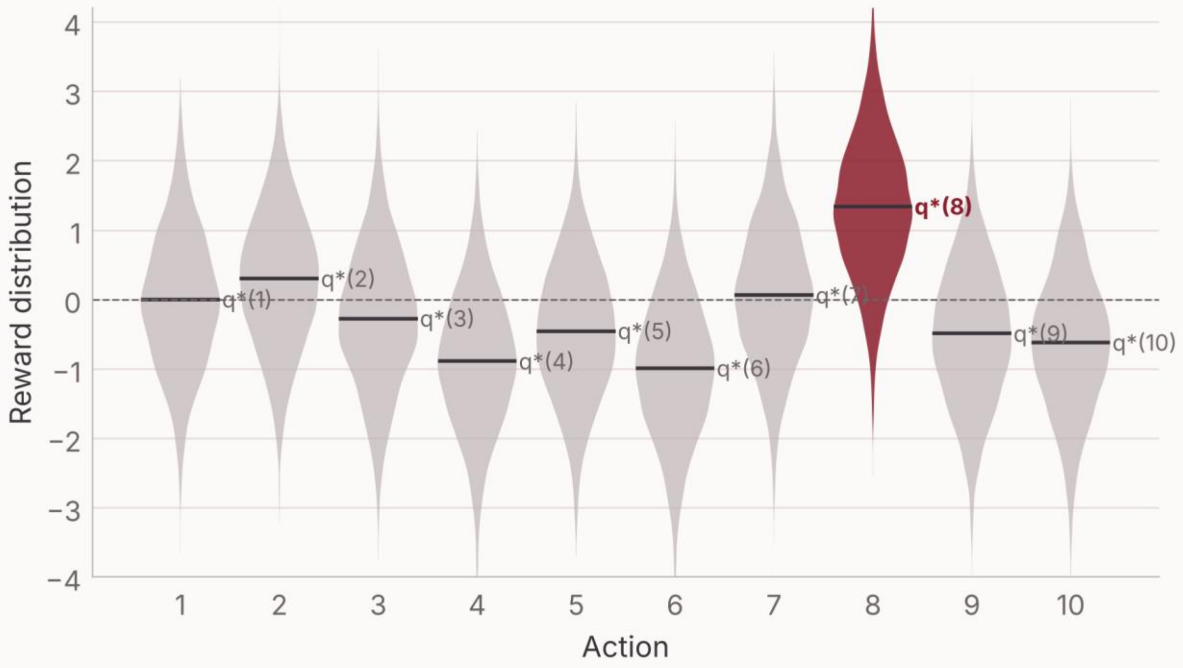
$a^* = \text{argmax}_a\, q^*(a)$: the one we hope to find

The agent never sees this chart

It only gets single rewards: one point drawn from a violin each time it tries an action.

One try per arm is not enough

A reward is typically within **±1** of $q^*(a)$: about as large as the **differences between the true values** of the arms. Each arm must be tried several times.



One problem drawn from the testbed: the optimal action here is 8

Three sources of randomness

The problem

Every run draws new values q*(a): some problems are easy, some hard

The rewards

Each reward is drawn around q*(a): the same action pays differently each time

The agent

Exploration steps and ties are random: two runs make different choices

SO ONE RUN PROVES NOTHING

A good method can be unlucky and a bad method lucky. To compare methods fairly we run **many independent runs**, each on a new problem, and **average** the results.

PART 1 · MEASURE

The experiment, step by step

ONE EXPERIMENT

```
for run = 1 to 2000:
    draw a new problem: $q^*(a) \sim N(0,1)$
    reset $Q(a)$ and $N(a)$ for all a
    for t = 1 to 1000:
        select $A_t$ with the method
        get $R_t$, update $Q(A_t)$
        record $R_t$ and $[A_t = a^*]$
```

Average reward at step t

mean of R_t over the 2000 runs: how much the method earns

% optimal action at step t

fraction of runs with $A_t = a^*$: how often it picks the best arm

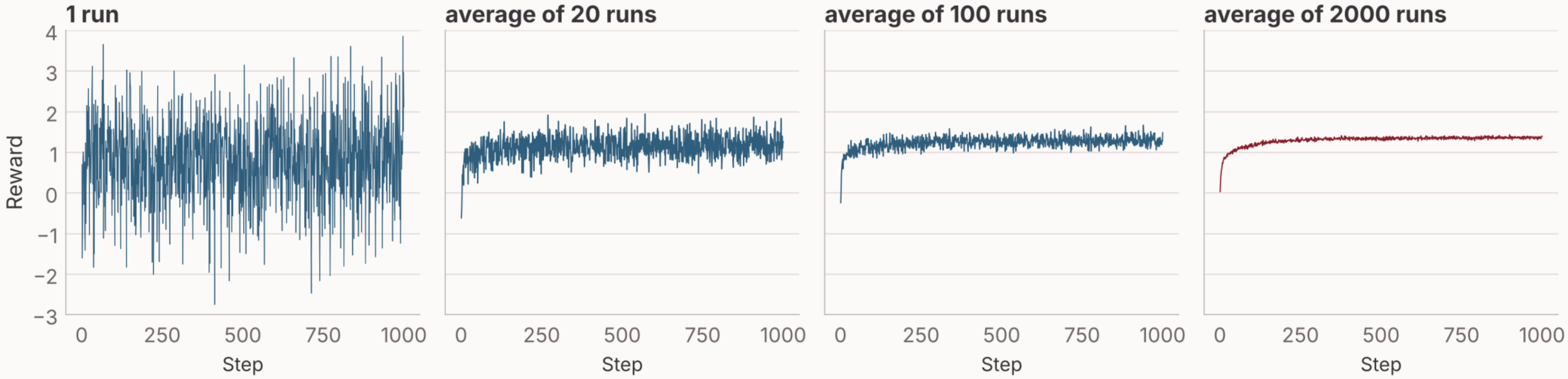
BEST POSSIBLE

Always picking a*: $E[\max_a q^*(a)] \approx$ **1.54** per step

Estimates: sample averages, $Q_1(a) = 0$, unless stated otherwise. 2000 × 1000 = 2 million steps per method.

One run is noise, many runs show the method

Reward received at each step by ε-greedy (ε = 0.1), **whatever action it chose** · averages taken step by step across runs



Averaging removes the noise

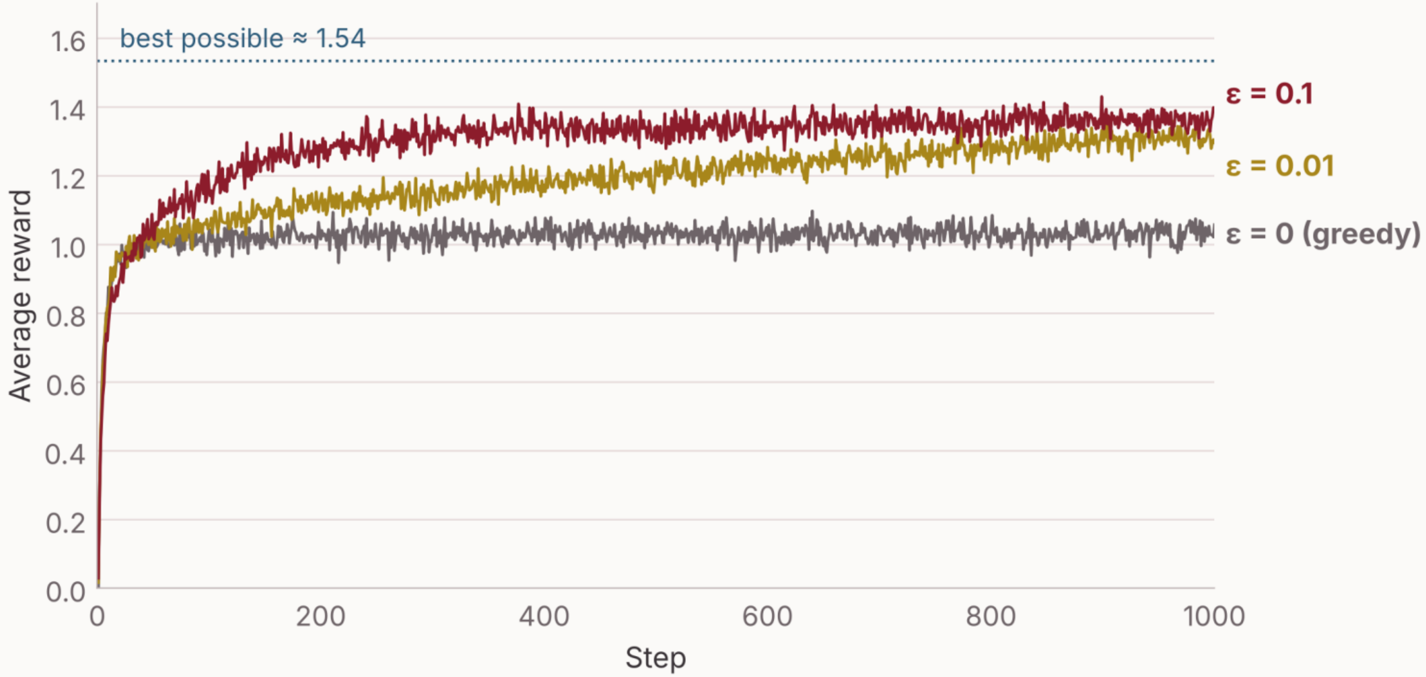
The spread shrinks as 1/√runs: from 1 run to 2000 the curve becomes smooth

What 2000 runs show

A fast rise in the first ~200 steps, then a slow climb towards a plateau

EXPERIMENT 1 · ε-GREEDY

Average reward: greedy gets stuck



Greedy · ε = 0

Fastest at the very start, then stuck at ≈ 1.0

ε = 0.1

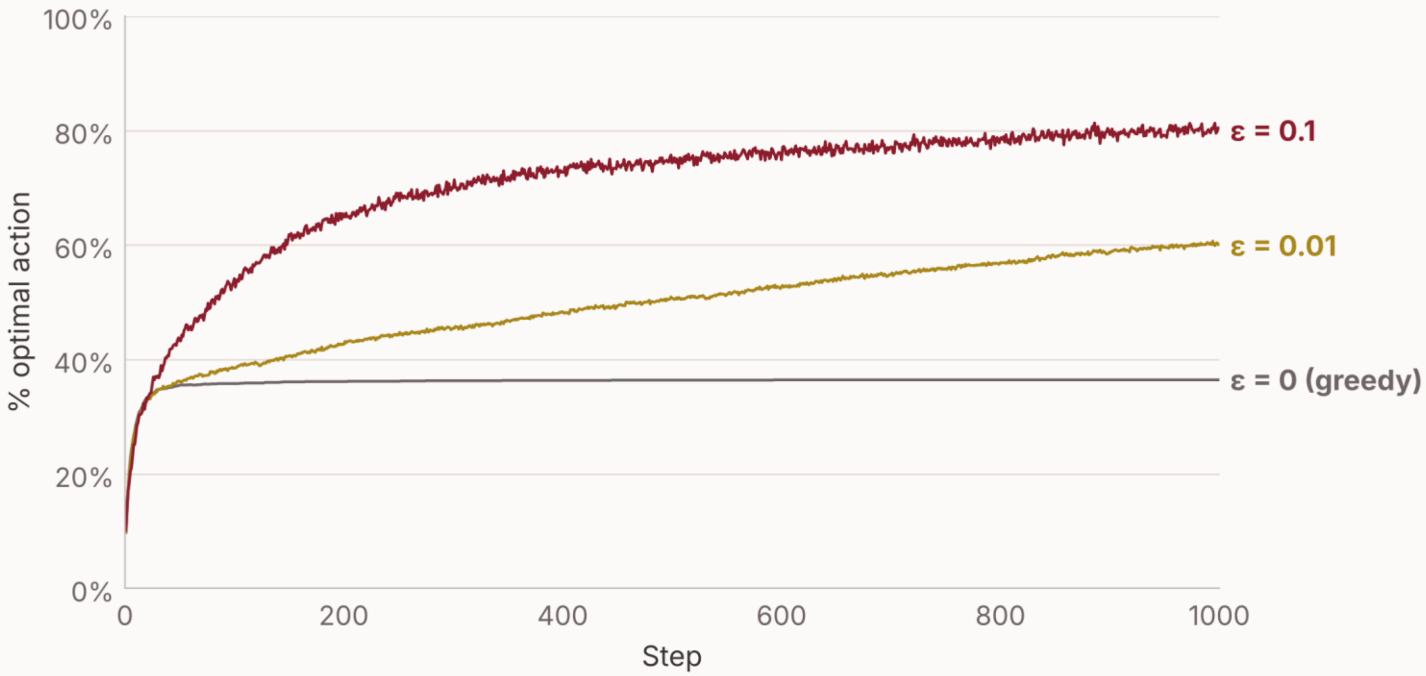
Explores more: learns fastest and reaches ≈ 1.35

ε = 0.01

Explores less: slower, but still climbing at step 1000

EXPERIMENT 1 · ε-GREEDY

% optimal action: exploring finds the best arm



Greedy finds a* in ⅓ of the runs

In the other ⅔ a bad first impression is never corrected

ε = 0.1 has a ceiling

At most 1 − ε + ε/k = 91%: it keeps exploring forever

ε = 0.01 is slow but steady

60% at step 1000, still rising

EXPERIMENT 1 · ε-GREEDY

No ε is best at every time

method	first steps	step 1000	in the long run
ε = 0 (greedy)	fastest	avg ≈ 1.03 · optimal 36%	**stuck on a suboptimal arm**
ε = 0.1	fast	avg ≈ 1.35 · optimal 80%	capped at 91% optimal
ε = 0.01	slow	avg ≈ 1.30 · optimal 60%	**best: up to 99.1% optimal**

Large ε

Learns fast, pays for exploration forever

Small ε

Learns slowly, wastes little in the long run

It depends on the task

Horizon, reward noise (more noise → more exploration) and nonstationarity

EXERCISE 1 · SUTTON & BARTO 2.3

In the long run, ε = 0.1 or ε = 0.01?

Run the testbed for a very long time, so that every estimate has converged: Q(a) = q*(a).

For ε = 0.1 and for ε = 0.01, compute:

(a) the probability of selecting the optimal action

(b) the expected reward per step

Which method is better, and by how much?

HINT

Greedy step: the greedy action is a*

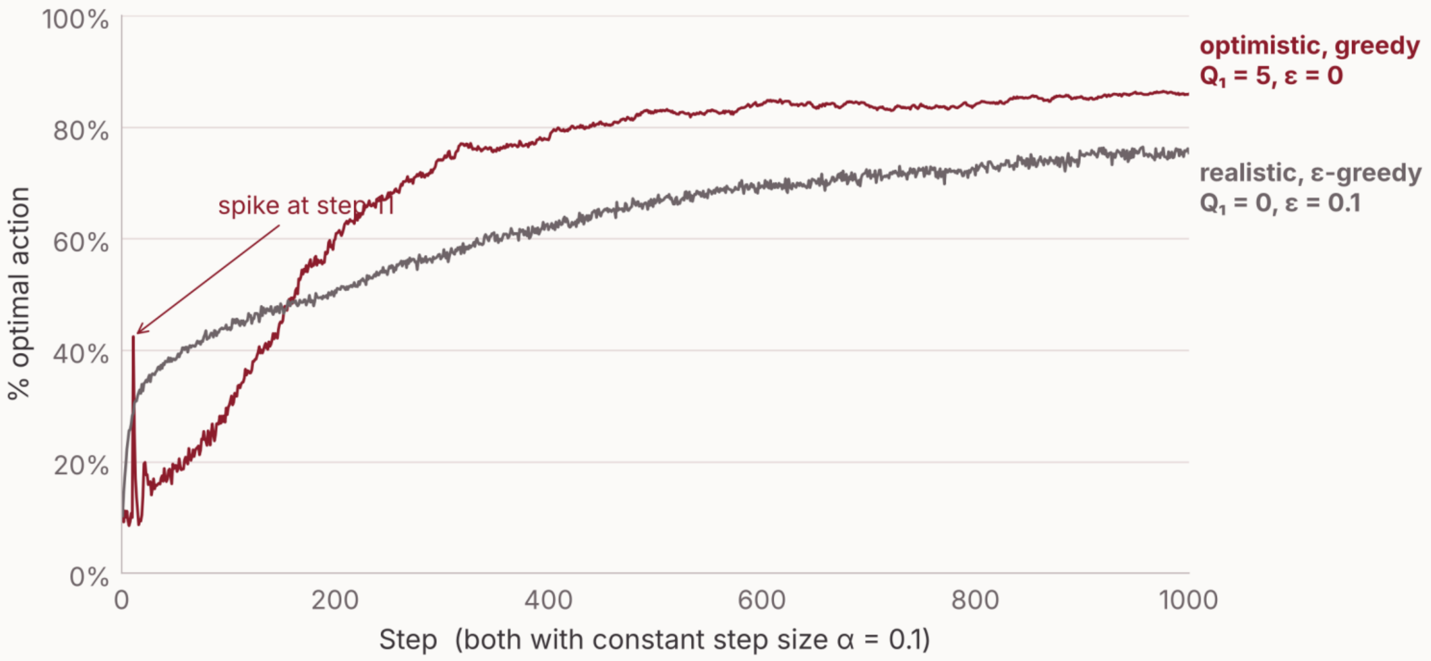
Random step: each of the k = 10 arms with probability 1/k

$E[\max_a q^*(a)] \approx 1.54$

average of $q^*(a)$ over the arms ≈ 0

EXPERIMENT 2 · OPTIMISTIC INITIAL VALUES

Optimism: worse at first, better later



Setup

Greedy with $Q_1 = 5$ vs. ε-greedy with $Q_1 = 0$, $\varepsilon = 0.1$; both $\alpha = 0.1$

At first: worse

Every action disappoints, so it keeps switching: lots of exploration

Later: better

Exploration fades by itself: ≈ 86% vs. 76% optimal at step 1000

PART 1 · WHAT WE LEARNED

ε-greedy vs. optimistic initial values

	ε-greedy	**optimistic initial values**
how it explores	random action w.p. ε	greedy, pushed by disappointment
when	always, a fraction ε of the steps	mostly at the beginning
to tune	ε (depends on the reward noise)	Q_1 (needs the scale of the rewards)
stationary problem	good, but pays ε forever	**very good**
changing problem	**good: never stops exploring**	**poor: the start happens only once**

Neither uses what the agent knows about its own **uncertainty**: which actions have been tried often, and which only a few times.

WHERE WE ARE

Next: explore where we are uncertain

Part 1 · Measure

- ✓ The 10-armed testbed
- ✓ ε-greedy on the testbed
- ✓ Optimistic initial values on the testbed

Part 2 · Explore smarter

- → **Uncertainty in the estimates**
- ○ Upper confidence bound (UCB)
- ○ UCB on the testbed, and in networks

Three approaches to exploration: **naive** (add noise: ε-greedy) · **optimistic initialisation** (assume the best until proven otherwise) · **optimism in the face of uncertainty**

ε-greedy explores blindly

action	Q(a)	N(a): times tried	what we know	ε-greedy exploration
a_1	**1.5**	200	good, well known	**greedy**
a_2	1.4	3	**looks good, very uncertain**	prob. $\varepsilon/4$
a_3	−2.0	150	clearly bad	**prob. $\varepsilon/4$: wasted**
a_4	0.1	2	**very uncertain**	prob. $\varepsilon/4$

Uniform exploration

A random step is as likely to re-test a_3, known to be bad, as to test a_2

Can we do better?

Explore the actions that are uncertain and could still be the best

UNCERTAINTY

Our estimates are uncertain

Q(a) is computed from a few noisy rewards: it is only a guess of q*(a), and some guesses are more reliable than others.

Easy problem

Two arms: one always good, one always bad, no noise.

Try each once and you are **done**: you know which is best.

Hard problem

Two arms with different means, but very noisy rewards.

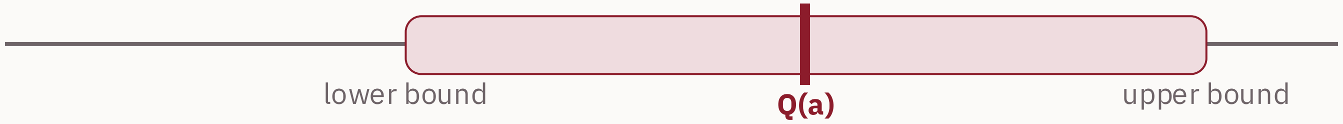
It takes **many tries** to tell them apart: similar-looking arms, different means.

THE KEY QUESTION

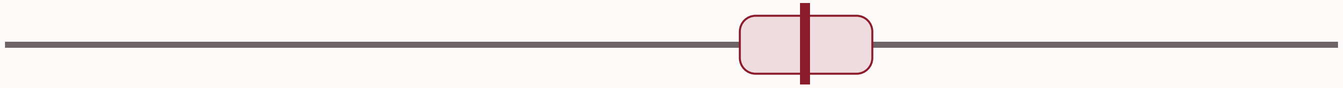
How much do we trust each estimate? We need a measure of uncertainty, not only the mean.

UNCERTAINTY

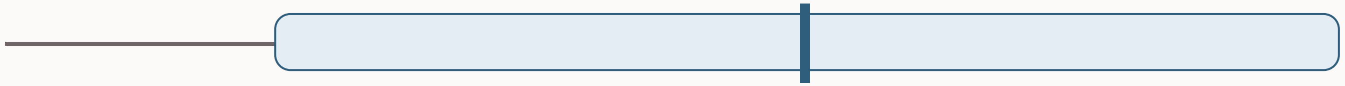
A confidence interval around each estimate



The box is a **confidence interval**: we are confident that the true value q*(a) lies somewhere inside it.



tried many times: small interval, we are **certain**



tried a few times: large interval, we are **uncertain**

How wide?

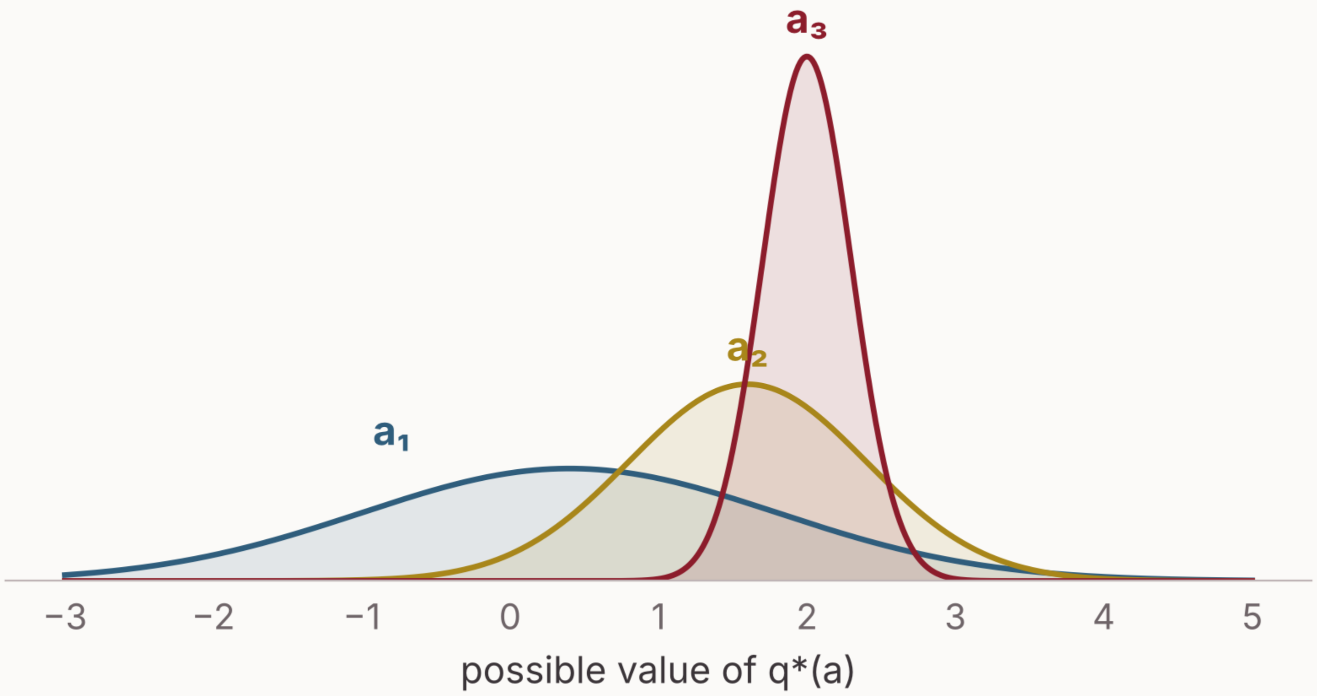
The width shrinks with the number of tries N(a), roughly as **$1/\sqrt{N(a)}$**

Halving the uncertainty takes 4× as many tries.

Uncertainty is **reduced by trying the action**: exploration buys information.

OPTIMISM IN THE FACE OF UNCERTAINTY

Which action should we pick?



What the agent believes about q*(a) after a few steps

a_3 (red)

Tried many times: narrow, best estimate

a_1 (blue)

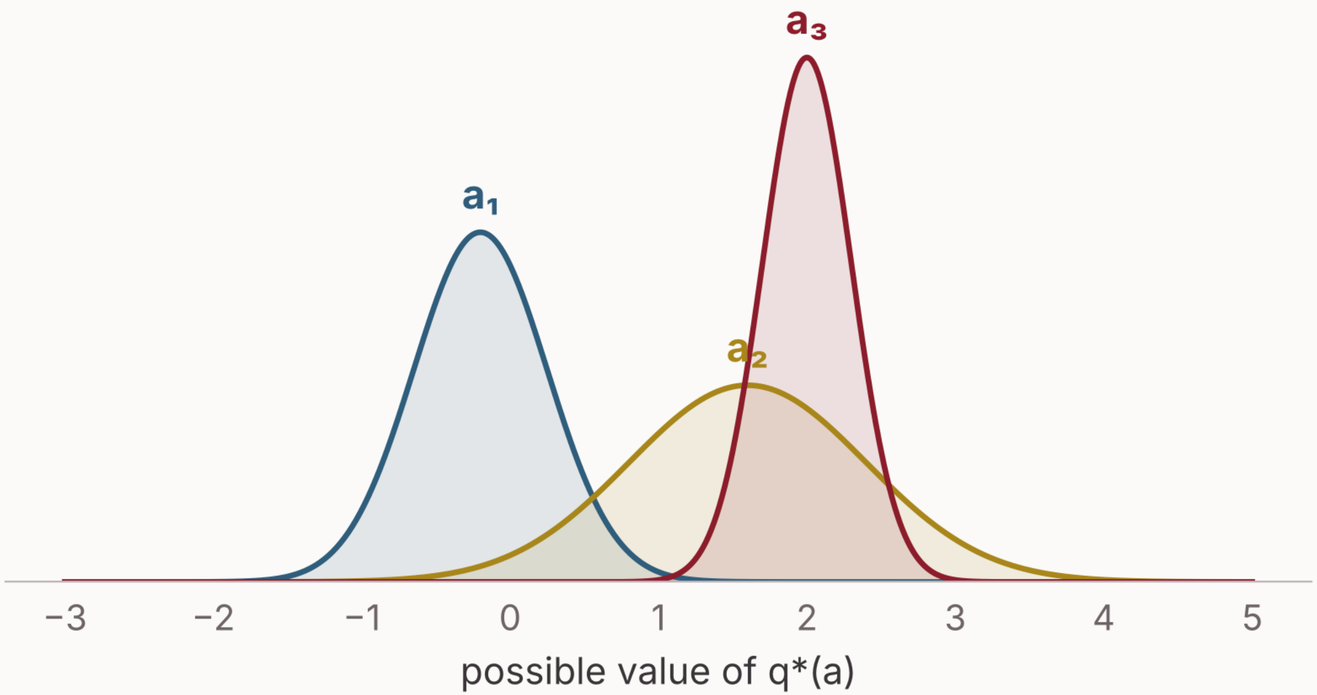
Tried a few times: wide, could be as high as 4

THE PRINCIPLE

Do not take the arm you believe is best: take the one that **could be the best**.

OPTIMISM IN THE FACE OF UNCERTAINTY

After trying a_1: less uncertain



After a few more tries of a_1: its curve is narrower, and its estimate fell

Either way, we win

a_1 is good: we found a better action and get high rewards

a_1 is bad: we learned it, and stop trying it

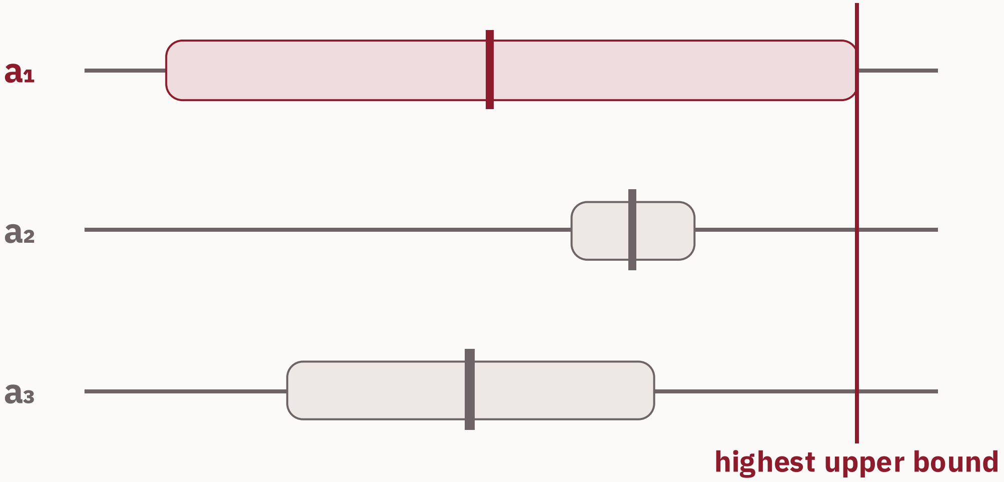
Then repeat

Pick the next most promising action; uncertainty shrinks where it matters, until we home in on the best action

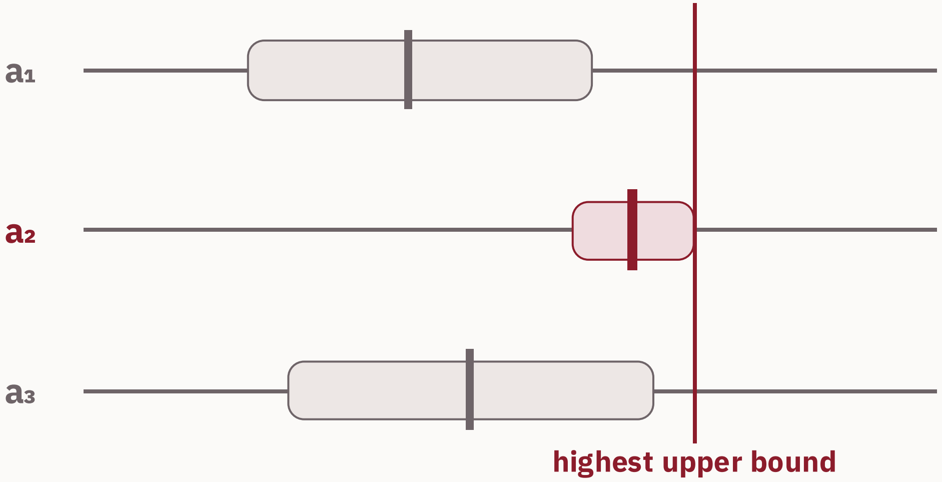
UPPER CONFIDENCE BOUND (UCB)

Pick the action with the highest upper bound

Step 1


a_1
a_2
a_3
highest upper bound

Step 2: after trying a_1


a_1
a_2
a_3
highest upper bound

Wide interval → explore

a_1 is chosen because we know little about it

High estimate → exploit

a_2 is chosen because its estimate is the highest

UPPER CONFIDENCE BOUND (UCB)

UCB action selection

$$A_t \doteq \underset{a}{\operatorname{argmax}} \left[\underbrace{Q_t(a)}_{\text{exploitation: estimated value}} + \underbrace{c\sqrt{\frac{\ln t}{N_t(a)}}}_{\text{exploration: uncertainty bonus}} \right]$$

$N_t(a)$

times a was selected before t: more tries → smaller bonus

ln t

grows, but slowly: an action left aside sees its bonus grow, and is tried again

c > 0

controls the amount of exploration: c = 0 is greedy

$N_t(a) = 0$

a counts as maximising: every action is tried once first

UPPER CONFIDENCE BOUND (UCB)

How large is the bonus?

At step $t = 10\,000$: $\ln t \approx 9.21$. Bonus = $c \cdot \sqrt{9.21 / N_t(a)}$

$N_t(a)$: times tried	$\sqrt{\ln t / N_t(a)}$	bonus	meaning
5 000	$\sqrt{9.21 / 5000}$	0.043 c	well known: almost no bonus
100	$\sqrt{9.21 / 100}$	**0.303 c**	**50× fewer tries → $\sqrt{50} \approx 7$× larger bonus**
10	$\sqrt{9.21 / 10}$	0.96 c	uncertain: large bonus
1	$\sqrt{9.21 / 1}$	**3.03 c**	**almost unknown: it will be tried soon**

Square root

The bonus falls slowly with the tries: 4× more tries halve it

Logarithm

Time raises every bonus, slowly: no action is abandoned forever

EXERCISE 2

Which action does UCB select?

After 10 steps the agent has these estimates and counts. It is now choosing the action at step $t = 11$.

action	Q(a)	N(a)
a_1	1.0	4
a_2	1.5	5
a_3	0.8	1

(a) greedy?

(b) UCB with $c = 2$?

(c) UCB with $c = 0.5$?

HINT

score = $Q(a) + c \cdot \sqrt{\ln t / N(a)}$

$\ln 11 \approx 2.40$

Compute the bonus of each action first.

The UCB algorithm

```
Initialize, for a = 1 to k:
    $Q(a) \leftarrow 0$,  $N(a) \leftarrow 0$
Loop, for t = 1, 2, …:
    if some $N(a) = 0$:  $A \leftarrow$ that a
    else:  $A \leftarrow \operatorname{argmax}_a [Q(a) + c \cdot \sqrt{\ln t / N(a)}]$
    $R \leftarrow \text{bandit}(A)$
    $N(A) \leftarrow N(A) + 1$
    $Q(A) \leftarrow Q(A) + (1/N(A)) \cdot [R - Q(A)]$
```

Only the selection changes

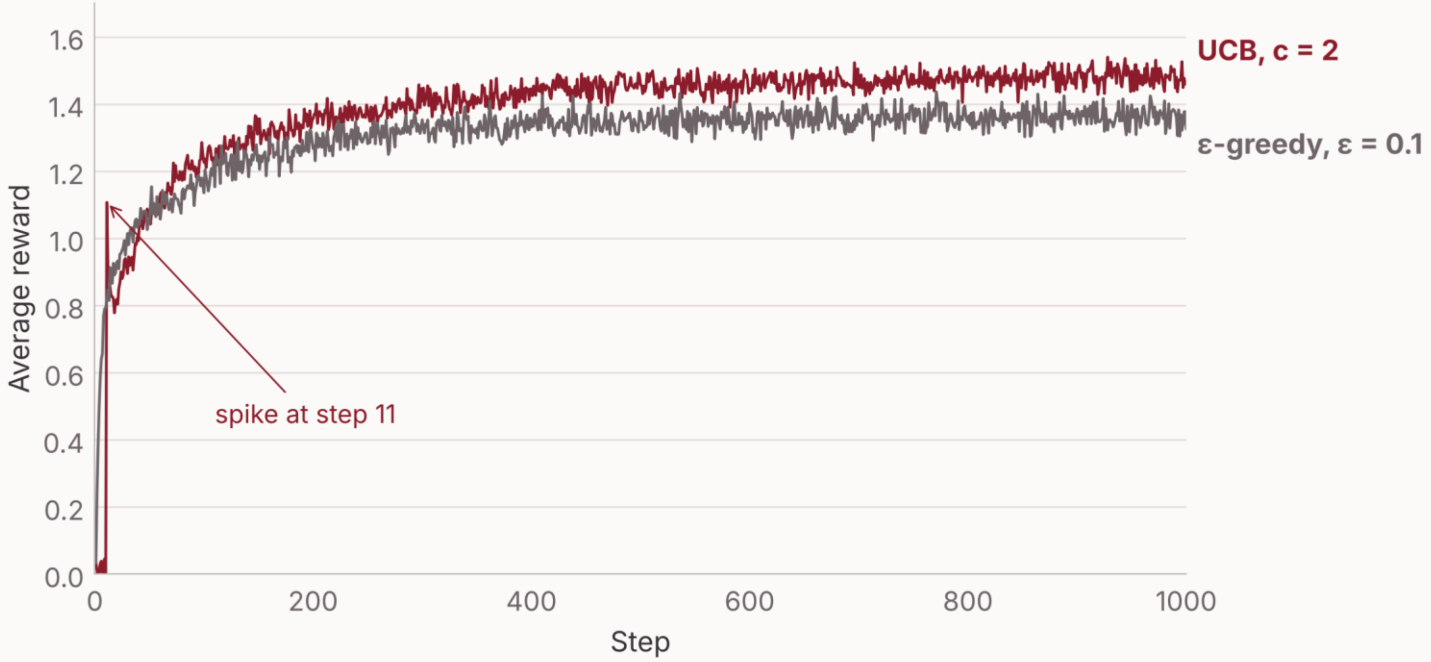
Highlighted lines: the rest is the algorithm of lecture 2

Cheap

Two numbers per action, $Q(a)$ and $N(a)$, plus one square root each step

UCB ON THE TESTBED

UCB beats ε-greedy



At first: explores more

Every action is tried; uncertain ones are retried systematically

Later: explores less

The bonuses shrink; ε-greedy still explores 10% of the time

Spike at step 11

All N(a) = 1, equal bonuses: UCB picks the best first reward, then moves on

COMPARING ALL METHODS

Which method when?

method	how it explores	stationary	changing problem
greedy	never	**gets stuck**	**gets stuck**
ε-greedy	at random, a fraction ε of the steps	good	good: explores forever
optimistic Q_1	at the start, by disappointment	very good	**poor**
UCB	where uncertainty is high	**best on the testbed**	**slow: counts never forget**

UCB and nonstationarity

Counts never forget; fix: count only recent tries (sliding-window, discounted UCB)

UCB and large problems

One count per action: hard with many actions, or with states

CONCEPT CHECK

True or false?

T / F ?	**1.** One run of a method on one problem is enough to compare it with another.
T / F ?	**2.** With $\varepsilon = 0.1$, ε-greedy picks the optimal action at most 91% of the time.
T / F ?	**3.** UCB always selects the action with the highest estimate $Q(a)$.
T / F ?	**4.** Optimistic initial values are well suited to nonstationary problems.

EXAM PRACTICE · EXERCISE 1

ε-greedy with sample averages

A 3-armed bandit uses ε-greedy with sample-average estimates, $Q_1(a) = 0$ for all a, ties broken at random. The first five steps are in the table.

(a) Give Q(a) for every action after each step.

(b) At which steps was the action certainly a random (exploration) step? At which steps could it have been either?

(c) With ε = 0.1, what is the probability of selecting action 3 at step 6?

t	**A_t**	**R_t**
1	1	2
2	2	1
3	1	0
4	3	3
5	3	1

EXAM PRACTICE · EXERCISE 2

Constant step size α

One arm, $Q_1 = 0$, constant step size $\alpha = 0.5$. The arm returns the rewards $R_1 = 4$, $R_2 = 0$, $R_3 = 2$, $R_4 = 2$.

(a) Compute Q_2, Q_3, Q_4, Q_5 with $Q_{n+1} = Q_n + \alpha[R_n - Q_n]$.

(b) Write Q_5 as a weighted sum of $Q_1, R_1, \ldots, R_4$. Check that the weights sum to 1.

(c) What would the sample average give? Which estimate would you prefer if the value of the arm changes over time, and why?

HINT

$Q_{n+1} = \alpha R_n + (1-\alpha)Q_n$

weight of R_i: $\alpha(1-\alpha)^{n-i}$

weight of Q_1: $(1-\alpha)^n$

EXAM PRACTICE · EXERCISE 3

Optimistic initial values by hand

Two arms with deterministic rewards: a_1 always gives $R = 1$, a_2 always gives $R = 2$. The agent is greedy ($\varepsilon = 0$), $\alpha = 0.5$, and ties are broken in favour of a_1.

(a) Trace the first 6 steps with $Q_1(a) = 5$ for both arms: chosen action and estimates.

(b) Repeat with $Q_1(a) = 0$.

(c) Which run finds the best arm? Why?

HINT

$Q \leftarrow Q + 0.5 \cdot (R - Q)$

Greedy picks the larger Q; on a tie, a_1.

EXAM PRACTICE · EXERCISE 4

UCB action selection

At step t = 21 the agent has these estimates and counts (ln 21 ≈ 3.04):

action	Q(a)	N(a)
a_1	0.6	10
a_2	0.9	8
a_3	0.4	2

(a) greedy?

(b) UCB, c = 1?

(c) UCB, c = 0.5?

(d) why do (b) and (c) differ?

HINT

score = Q(a) + c·√(ln t / N(a))

Compute √(ln t / N(a)) once, then reuse it for both values of c.